SF

SMARTFUEL

COMMERCIAL PROTOTYPE PROPOSAL

CAMERA / OCR / DATA / DECISION

AI-ENABLED ANPR & CNG HYDROTESTING VERIFICATION

One dashboard. Automatic number recognition.
Clear YES / NO demo decision with reasons.

7 DAYS
Software target

+7 DAYS
Optional government API

Prepared for client discussion | October 2026



One software. Two data modes.

The complete inspection process remains the same whether records come from a local demo database or an authorised government integration.

MODE A / DEMONSTRATION DATABASE

100 fictional vehicle records, stored CNG endorsement and hydrotesting fields, sample certificates, and local decision rules.

MODE B / AUTHORISED API

Use the same camera, OCR and dashboard; replace the data lookup with approved vehicle and hydrotesting APIs after credentials are provided.

CLIENT EXPERIENCE DOES NOT CHANGE

Camera / Upload → Plate text → Vehicle + cylinder checks → Decision → Report + entry log

Data mode must be disclosed: demonstration eligibility is not an official permission to fuel.

The client requirements, mapped to the build

Mandatory functions from the supplied CNG hydrotesting Scope of Work and the proposed pilot interface.

01	**Identify number plate**	USB camera / image upload + ANPR OCR	CORE
02	**Check CNG endorsement on RC**	Stored demo RC record / future approved API	CORE
03	**Count installed cylinders**	Vehicle-to-cylinder registry	CORE
04	**Check each hydrotesting validity**	Expiry and evidence rules per cylinder	CORE
05	**Show invalid or absent certificate**	Separate CNG station decision screen	CORE
06	**Display / download certificate**	Sample certificate + report PDF	CORE
07	**Admin dashboard + role access**	Station users, logs, audit and reports	PILOT

Step-by-step: car entry to the final decision

An end-to-end journey the client can watch on one software screen and a separate station display.

1 CAR ARRIVES

An authorised camera watches the entry area.

2 CAPTURE

Automatic USB frame or Upload Image button.

3 IMAGE → TEXT

OCR reads plate; low-confidence scans are reviewed.

4 SAVE EVENT

Record image, station ID and server timestamp.

5 DATA LOOKUP

Match registration in your own database or API.

6 COMPLIANCE

RC CNG endorsement + cylinder hydrotest validity.

7 YES / NO / REVIEW

Show recommendation and specific reasons.

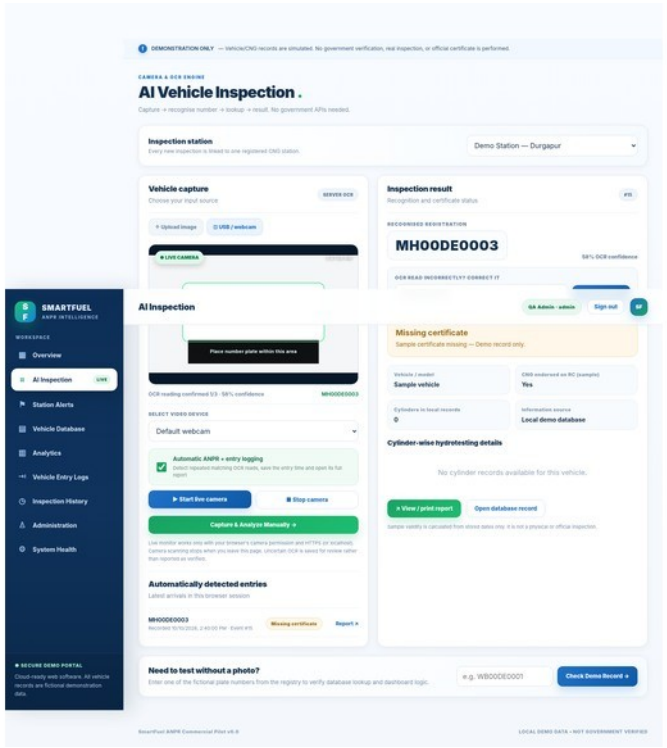
8 LOG + REPORT

Save history, certificate view and PDF report.

Automatic webcam monitoring is a prototype observation; reliable physical gate crossing needs on-site validation.

Step 1–2 / Camera monitoring and image capture

Two ways to supply a vehicle image. A normal USB webcam runs on the operator PC and submits frames securely to the software.



SOFTWARE SCREEN PREVIEW • SIMULATED CAMERA

OPTION 1 / LIVE USB WEBCAM

On-screen feed; periodic OCR sampling and automatic entry logging.

OPTION 2 / IMAGE UPLOAD

Choose an existing vehicle photo and analyze on demand.

IMPORTANT TECHNICAL DETAIL

USB camera connects to the local operator computer, not directly to the remote KVM2 VPS. Browser capture requires HTTPS and permission.

Step 3 / Image becomes a registration number

OCR extracts characters; the operator may confirm or correct uncertain readings before compliance is assessed.



FICTIONAL SAMPLE PLATE

OCR OUTPUT

WB00DE0001

DEMO DATA

MANUAL CONFIRMATION AVAILABLE

Plate text is normalized, checked for basic format and submitted to the data lookup service.

Low-confidence OCR must remain REVIEW until the plate is confirmed; AI should not guess eligibility.

Step 4–6 / The two compliance checks

The software evaluates the recognized registration against stored records; the same rules apply after approved API integration.

CHECK A / CNG ON RC

Is CNG endorsed in the matching RC record?

YES → Continue to cylinder inspection.
NO → Show HOLD with RC-related reason.
UNKNOWN → Show REVIEW.

CHECK B / HYDROTEST VALIDITY

How many cylinders are installed? Is each test record present and inside its validity period?

ALL PASS → Demo YES.
EXPIRED / MISSING → HOLD or REVIEW with exact reason.

DECISION RULE

YES only when both checks pass • NO / HOLD for failed records • REVIEW when data is missing or OCR is uncertain.

Step 7 / A separate display for the station

After each accepted inspection, the second monitor updates automatically with the plate and the exact demo decision reason.

DEMO RESULT: NO / HOLD

WB00DE0001

Reason for holding CNG filling:

Cylinder hydrotest validity has expired in the sample records. Operator review required.

Entry observation: saved • Inspection report: available

DEMO RESULT: YES

DL00DE0002

RC CNG endorsement: YES

All stored cylinder hydrotest checks pass in the demonstration records.

Entry observation: saved • Sample certificate: available

These are demo software recommendations only. Actual CNG supply requires approved data and station procedures.

Step 4 + 8 / Entry time, image and report history

Each accepted camera reading creates a traceable inspection event; manual uploads are recorded as inspections, not physical arrivals.

1

ENTRY TIME

Server-side timestamp and station

2

VEHICLE

Recognized and confirmed plate

3

EVIDENCE

Captured image + OCR confidence

4

OUTCOME

RC + cylinder check and reason

5

EXPORT

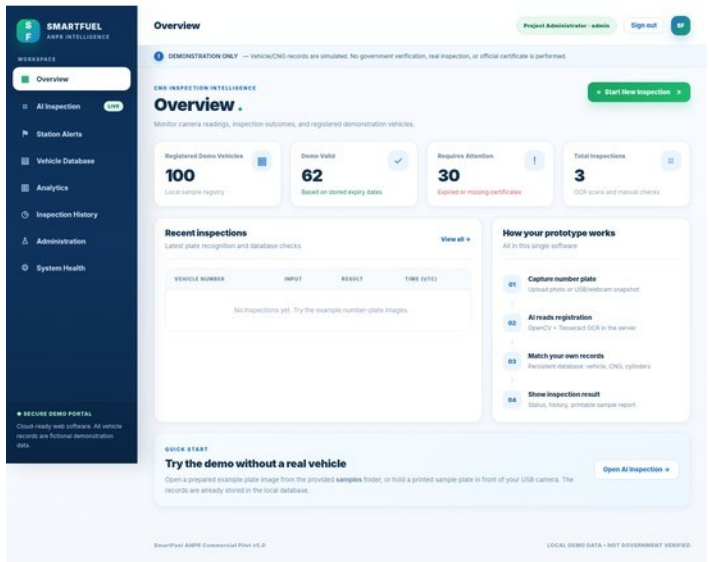
Inspection PDF and CSV history

WHAT THE OPERATOR CAN REVIEW

- Search by plate / station / date
- Open a single-vehicle inspection report
- Review the exact reason and the recorded certificate evidence

Latest software interface previews

These preview images were produced with simulated/demo API responses; they show the proposed UI, not a live hosted system.



Operations dashboard

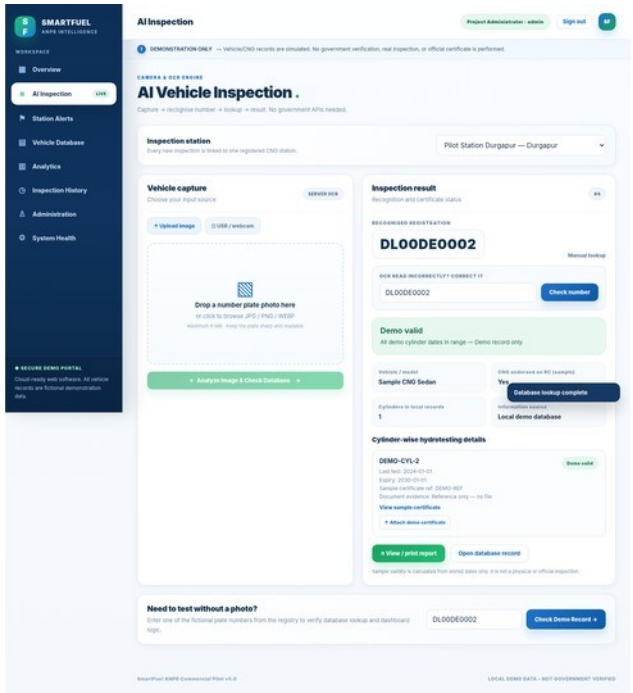
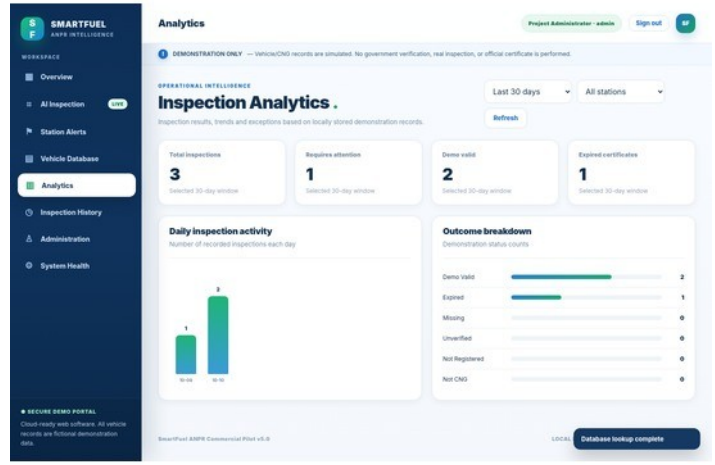


Image upload + inspection



Analytics & reporting

How the software will run on Hostinger KVM2

Actual v7 technology stack: one secure browser application plus a Python OCR and verification backend.

OPERATOR

Browser + USB camera / upload

→

FRONTEND

HTML / CSS / JavaScript

→

BACKEND

Python FastAPI + OpenCV / Tesseract

→

DATABASE

PostgreSQL + file storage

HOSTING AND DATA MODE

Ubuntu + Docker + Caddy HTTPS • Hostinger KVM2 VPS
Demo phase: PostgreSQL local record lookup | Optional phase: authorised API adapter

Controls needed for a credible commercial pilot

A usable and accountable prototype must do more than read a number plate.

ACCESS

Admin, operator and viewer permissions; secure login.

CAMERAS

Approved operator devices, permissions and protected snapshots.

RECORDS

Vehicle logs, evidence links, inspection history and audit trail.

PRIVACY

Avoid showing unnecessary personal data on public station screens.

RECOVERY

Database backup, application health checks and restore practice.

DISCLOSURE

Identify simulated records separately from official/authorised data.

Week 1 / Complete the software prototype

Seven-day target for finishing and validating the source-built workflow on the selected hosting environment.

DAY 1	Finalise requirements, database, and approval states
DAY 2	USB webcam feed + Upload Image workflow
DAY 3	OCR normalization + manual confirmation
DAY 4	RC and cylinder rules + YES / NO / REVIEW
DAY 5	Station decision display + timestamps + reports
DAY 6	Hostinger KVM2 deployment + security basics
DAY 7	End-to-end demonstration and acceptance tests

Week 2 / Optional authorised government API

Seven-day integration target starts only after approved credentials, endpoint documentation, permissions and sample responses are available.

DAY 8	API credentials / access validation
DAY 9	Government fields → software database mapping
DAY 10	RC CNG endorsement API integration
DAY 11	Cylinder count / hydrotesting source integration
DAY 12	Error, missing-data and certificate handling
DAY 13	Secure logging, permissions and failover tests
DAY 14	End-to-end client verification and sign-off

Transparent project investment: two options

Only API integration changes the development price; the demonstration workflow, dashboard and user steps remain the same.

Hostinger VPS KVM2	₹13,154.64
Software development	₹50,000.00
Development tools	₹9,100.00
Two domains: test + host	₹2,000.00
Optional authorised government API development	**+ ₹30,000.00**

OPTION A / DEMO DATA

₹74,254.64

1 week software target

OPTION B / GOV API

₹1,04,254.64

Week 1 + optional Week 2

Figures supplied by the project team. API licensing/access charges, physical cameras, taxes (if applicable) and 24×7 service costs are not included unless separately agreed.

What the client can verify during the live demo

A tangible acceptance checklist rather than a design-only presentation.

- **A** Launch the camera monitor / upload a sample vehicle image
- **B** Watch OCR generate and confirm a number plate
- **C** Find the matching record in the software database
- **D** Check CNG on RC and each cylinder hydrotesting record
- **E** See YES / NO / REVIEW + precise reason on station display
- **F** Retrieve captured entry time, image, report and log entry

Acceptance of official compliance verification is separate from acceptance of the demonstration software.

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SMARTFUEL / CLIENT PROPOSAL

Ready to demonstrate the complete decision workflow.

SOFTWARE BUILD 7 DAYS	OPTIONAL API +7 DAYS

Same camera, OCR, checks, decision display, entry logs and reports in both data modes.

Next: approve scope, identify USB camera / station, confirm KVM2 access and obtain API approvals if required.